

the wind pressure and exposed area. The wind shall be assumed horizontal, except as otherwise specified in Article 3.8.2, and can come from any horizontal direction. Areas that do not contribute to the extreme force effect under consideration may be neglected in the analysis.

For typical bridges, wind loads on the substructure from the superstructure may be determined for the wind in the direction transverse to the bridge in elevation then adjusted for various angles of attack using the provisions of Article 3.8.1.2.3a.

3.8.1.1.2—Wind Speed

The design 3-second gust wind speed, V , used in the determination of design wind loads on bridges shall be determined from Figure 3.8.1.1.2-1. For areas designated as a special wind region in Figure 3.8.1.1.2-1, the Owner shall approve the 3-second gust wind speed.

The wind speed shall be increased where records, experience, or site-specific wind studies indicate that wind speeds higher than those reflected in Figure 3.8.1.1.2-1, based upon seven percent probability of exceedance in 50 years, are possible at the bridge location.

Unless a site-specific wind study is performed, wind speeds used for different load combinations shall be taken from Table 3.8.1.1.2-1.

Table 3.8.1.1.2-1—Design 3-Second Gust Wind Speed for Different Load Combinations, V

Load Combination	3-Second Gust Wind Speed (mph), V
Strength III	Wind speed, taken from Figure 3.8.1.1.2-1
Strength V	80
Service I	70
Service IV	0.75 of the speed used for the Strength III limit state

long bridges or when the substructure is unusually tall which may warrant using different structure heights in determining the wind pressure on different portions of the superstructure or substructure.

C3.8.1.1.2

Previous editions of these Specifications were based on fastest-mile wind speed. Since that criteria was based on distance, the effect was to average the wind speed over different lengths of time. The fastest-mile wind speed is no longer utilized by modern wind codes and the provisions herein are based on 3-second gust wind speed which means that the wind speed is averaged over 3 seconds.

Figure 3.8.1.1.2-1 shows the reference 3-second gust wind speed, at an elevation of 33.0 ft, for wind exposure Category C, as defined in Article 3.8.1.1.5, with a mean recurrence interval (MRI) of 700 years. The figure is taken from ASCE 7-10 (2010).

The wind pressure determined using the design 3-second gust wind speed with a load factor of 1.0 replaces the wind pressure determined in past specifications using fastest-mile wind speed with a MRI of 100 years and a load factor of 1.4. This is the reason the load factor for wind load on the structure, WS , in the Strength III load combination has been reduced from the 1.4 used in earlier editions of the specifications to the 1.0 currently shown. The change in the load factor was instituted when the design wind speed was changed from the fastest-mile wind used earlier to the design 3-second gust wind speeds shown in Figure 3.8.1.1.2-1.

The basis for the wind speeds specified for Strength V, Service I, and Service IV limit states is presented in Article C3.4.1.

For buildings and other structures, local building officials typically determine the 3-second gust wind speed to be used in the special wind regions under their jurisdiction. Bridge Owners will need to develop their own policy for the 3-second gust wind speed to be used in these regions.